MILITARY SPECIFICATION

MATTRESS, PNEUMATIC, (INSULATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of pneumatic mattress in one size and two classes.

1.2 Classification. Pneumatic mattresses may be either of the following classes.

Class 1 - Cemented seams, I - beam construction

Class 2 - Molded seams, C - beam construction

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-L-61	Laces, Nylon.
V-T-295	Thread, Nylon.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

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MIL-R-3065	Rubber, Fabricated Products.
MIL-C-7350	Cloth, Parachute, Nylon.
MIL-B-41826	Batting, Synthetic Fibers: Polyester, (Quilted and Unquilted.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-43659	Stopper, Inflating Tube, Pneumatic Mattress.
MIL-I-45208	Inspection Systems Requirements.

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods.
FED-STD-751	Stitches, Seams, and Stitchings.

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads for 40" X 48" Pallets.
MIL-STD-417	Classification System and Tests for Solid Elastomeric Materials (Symbols and Tests).

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

2-3-97	Mattress, Pneumatic; Construction Details.
2-3-99	Mattress, Pneumatic; Inflating Tube and Stopper.

(Figures 1 and 2 are miniature reproductions of the referenced drawings and are attached for information only.)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

	ASTM D 903	Peel or Stripping Strength of Adhesive Bonds.
*	ASTM D 2000	Rubber Products in Automotive Application.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. Requirements

3.1 Material.

3.1.1 Base cloth. The base cloth shall be plain weave, white nylon cloth made from 70 denier (8 tex), multifilament, semi-dull, nylon yarn having sufficient twist or other yarn preparation to allow the cloth to meet both the base cloth and coated cloth requirements. The cloth shall not be calendered, shall be heat set to prevent distortion, and shall conform to the physical and chemical requirements in table I when tested as specified in 4.3.1. As an alternate, nylon parachute cloth conforming to the weight, texture, and breaking strength requirements of type I of MIL-C-7350 may be used providing it meets all the other requirements cited in table I herein.

TABLE I. Base cloth requirements.

	Minimum	Maximum
Weight, ounces per square yard	2.0 (68 g/m^2)	2.3 (78 g/m^2)
Yarns per inch:		
- Warp	106 (42 yards/cm)	-
- Filling	92 (36 yarns/cm)	-
Breaking strength, pounds:		
- Warp	80 (356 N)	-
- Filling	60 (267 N)	-
Tearing strength, pounds:		
- Warp	3.5 (16 N)	-
- Filling	3.0 (13 N)	-
Shrinkage, percent:		
- Warp	-	2.0
- Filling	-	1.5
Chloroform-soluble material, percent	-	1.0
pH value	5.0	8.5
Copper content, percent	-	0.003
Manganese content, percent	-	0.0015

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3.1.2 Coating compound. The compound used for coating the cloth shall be a heat-vulcanizable chloroprene synthetic rubber containing not less than 60 percent chloroprene by volume. The remainder of the compound shall consist of softeners curing agents antioxidants pigments and reinforcing materials.

3.1.3 Coated cloth.

3.1.3.1 Top and bottom panels. The coating compound specified in 3.1.2 shall be applied to both sides of the base cloth in a quantity to meet the overall weight requirements in table II and shall be distributed so that one side shall have 1 1/2 ± 1/4 ounces per square yard (51 ± 9 g/m^2) and the other side shall have the remainder. The coating on the heavily coated side, which is to be on the inside of the mattress shall be colored black throughout and the surface color of the lightly coated side shall approximate Olive Green 207 (see 6.3). The coating shall be properly vulcanized and shall be free from defects which may affect serviceability. The olive green surface shall have a dull finish produced by an inert mineral dusting powder. The fully vulcanized coated cloth shall conform to the requirements in table II when tested as specified in 4.3.1.

TABLE II. Properties of vulcanized coated cloth.

Characteristic	Requirement		
	Minimum		Maximum
Overall weight, ounces per square yard	8.0	(271 g/m^2)	12.0
Breaking strength, pounds:			
- Warp	80	(356 N)	-
- Filling	65	(289 N)	-
Tearing strength, grams:			
- Warp	640	(6 N)	-
- Filling	512	(5 N)	-
Hydrostatic resistance:			
- Initial	1/		-
- After accelerated aging	1/		-
- After weatherometer	1/		-
- After strength of coating	1/		-
Adhesion, coating to cloth, pounds per 2 inch width			
- Heavily coated side	6.0	(27 N/51 mm)	-
- Lightly coated side	6.0	(27 N/51 mm)	-

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TABLE II. Properties of vulcanized coated cloth - Continued.

Characteristic	Requirement	
	Minimum	Maximum
Stiffness at 70°F (21°C), warp, mm	-	80
Stiffness at -40°F (-40°C), warp, mm	-	120
Abrasion (olive green side only), 20 cycles	2/	-
Blocking, 220°F (104°C), 30 minutes, scale rating	-	No. 1
Resistance to cold crack at -50°F (-46°C)	3/	-
Resistance to accelerated aging	3/ 4/	-

1/ There shall be no leakage.

2/ Visible loose fibers of the base cloth shall not be exposed in the center 1 inch (25 mm) on the abraded portion. Visible loose fibers shall be regarded as fibers from the yarns used to weave the base cloth and not as entire yarns or yarn plies.

3/ There shall be no cracking, flaking, or peeling of the coating.

4/ Specimen shall not become stiff and brittle, nor soft and tacky, when compared to an unexposed specimen of the same sample unit.

3.1.3.2 Beam material. Coated cloth for beams may be that specified in 3.1.3.1 or may be the same base cloth coated on one side with black coating compound specified in 3.1.2. The single coated cloth shall be properly vulcanized, shall have an overall weight between 6.5 and 8.5 ounces per square yard (220 and 290 g/m^2) and shall conform to the requirements for adhesion of coating to cloth as specified in table II. The beam material shall be tested as specified in 4.3.1.

3.1.3.3 Hinge material. Coated cloth for hinges used for reinforcing the fin edge of mattresses having the I-beam construction shall be the base cloth specified in 3.1.1, coated on one side with not less than 2.0 ounces per square yard (68 g/m^2) of the coating compound specified in 3.1.2. A light coating of a suitable rubber compound may be applied to the reverse side of the cloth, if desired, as a matter of convenience in assembly of the mattress.

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3.1.4 Cement. The compound use for cementing shall be synthetic or natural rubber cement, completely compatible with the coating on the cloth and shall have all other characteristics equal to the requirements for the coatings. The cement shall be such that the cementing of the batting meets the requirements in 3.3.3, the mattress meets the inflation test requirements in 3.4, and the beams and seams meet the adhesion requirements in 3.9.

3.1.4.1 Cement for bonding inflating tube. The cement shall be a natural or synthetic rubber cement, similar in composition to the coating compound and when tested as specified in 4.3.1 shall have a minimum peel strength of 4.0 pounds per inch of width (70 N/mm of width).

3.1.5 Thread. The thread shall be nylon thread, type I or II, class 1 or 2, size B, natural color or type III, size B, natural color conforming to V-T-295.

* 3.1.6 Inflating tube and stopper. The inflating tube shall be made of black rubber material meeting the requirements of type, class, grade AA615 or BC615 of MIL-R-3065 and ASTM D-2000. The tube shall be of the size and design shown in drawing 2-3-99. The stopper shall conform to MIL-S-43659.

3.1.7 Braid, nylon. The round braid shall conform to type II, class 1 of V-L-61 except that the ends shall have a plastic tip 1/2 inch (13 mm) in length (minimum) to prevent raveling. The braid shall be 12 ± 1/2 inches (305 ± 13 mm) long and shall be dyed black.

3.1.8 Batting, polyester. The polyester batting shall conform to the requirements for type III of MIL-B-41826 except the batting shall weigh not less than 5.0 (170 g/m^2) nor more than 5.5 ounces per square yard (186 g/m^2) and shall have a minimum thickness of 0.730 inches (18.5 mm). All launderability requirements shall not be applicable.

3.2 Design. The mattress shall conform in all respects to the design and dimensions shown on drawing 2-3-97 except that the insulating batting shall be included. Dimensions shall be measured in uninflated condition with tolerances of plus or minus 1/4 inch (6 mm) except where other tolerances are specified. The inflating tube shall be attached to the top panel at the upper or head end of the mattress in the air channel shown on drawing 2-3-97. The center of the inflating tube shall be located 4 ± 1 inch (102 ± 25 mm) from the head edge of the mattress. The inflating tube shall not be cemented over any beam flange.

3.3 Construction.

a. The mattress shall be constructed with one piece top and bottom panels of the coated cloth specified in 3.1.3.1, assembled with the heavily coated side of the cloth on the inside of the air chamber.

b. A piece of batting specified in 3.1.8, 3 1/4 ± 1/4 inches (83 ± 6 mm) wide by 62 ± 1 inches long (1575 ± 25 mm) shall be incorporated into each of the four center channels in such a manner as to finish 6 ± 1 inch (152 ± 25 mm) from each end of the mattress.

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c. A piece of batting specified in 3.1.8, 5 3/4 ± 1/4 inches (146 ± 6 mm) wide by 62 ± 1 inches long (1575 ± 25 mm) long shall be cut to the contour of the side channel and incorporated into the side channel in such a manner as to finish 6 ± 1 inches (152 ± 25 mm) from each end of the mattress.

d. The opening in the top panel of the mattress through which the inflating tube is to be inserted shall be 1 ± 1/8 inch (25 ± 3 mm) in diameter.

e. When cementing the inflation tube to the inside of the top panel, the mating surfaces shall be thoroughly cleaned.

f. The entire mating surface of the inflating tube shall be buffed and slightly abraded; and the mating surface of the coated fabric shall be cleaned to remove dusting powder and dirt.

g. The inflating tube shall be securely fastened by full cementing and vulcanizing by means of a press to the underside of the top mattress panel of the upper or head end of the mattress. The bottom mattress panel shall be protected by a circular adhered chafing patch made of either of the coated fabrics specified in 3.1.3.1 or 3.1.2.1 not less than 3 inches (76 mm) in diameter firmly affixed to the inside bottom mattress panel opposite the tube protrusion. The inflating tube shall be clean and free from flash.

h. The braid for securing the inflating tube stopper shall be inserted through the holes of the tube stem and stopper, each end looped around the hole and securely tied. The braid shall be 12 ± 1/2 inches (305 ± 13 mm) in length and when attached to the stopper, the stopper shall hang not less than 4 (102 mm) not more than 5 inches (127 mm) from the inflating tube hole. Either I-beam (class 1) or C-beam (class-2) construction of the mattress may be employed.

3.3.1 I-beam construction (class 1). The fin edge and I-beam seams shall be cemented full width and rolled to secure good adhesion. The I-beams shall be constructed from two strips of either of the coated specified in 3.1.3.1 or 3.1.3.2, not less than 2 1/2 inches (64 mm) wide, secured to each other by stitching or cementing. If cemented, the heavily coated sides of the cloth in each ply shall adhere one to the other. The effective height of the partition shall be not less than 1 1/8 inches (29 mm). If plies are joined by two rows of stitching, seam type SSa-2 with six to eight stitches per inch (2.4 to 3.1 stitches/cm) and stitch type 401 and 301 conforming to FED-STD-751 shall be used. The rows of stitching shall be at least 5/8 inches (16 mm) from the edges of the strip to permit attaching a flange to the body material having arms at least 5/8 inch (16 mm) on both sides. The beginnings and ends of all stitching shall be backstitched or tacked for at least 1/2 inch (13 mm). All breaks in stitching shall be overstitched for at least 1 inch (25 mm). The distance between rows of stitching shall be not less than 1 1/8 inches (29 mm). If plies are joined by cementing, they shall be full-cemented not less than

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1 1/8 inches (29 mm) wide along the center-line of the strips, leaving at least 5/8 inch (16 mm) along each edge for forming the flange top and bottom. The line of attachment of the arms to top and bottom panels shall be not less than 1/2 inch (13 mm) in width. The I-beams shall be of such length that when cemented to the body of the mattress, they shall be 1 1/8 ± 3/8 inches (29 ± 10 mm) from the fold of the hinge strip at each end of the mattress, except that the two outer beams shall be 3 1/2 ± 1/2 inches (89 ± 13 mm) from the fold at the head end. The coated cloth may be vulcanized either before or after assembly of the mattress. The two outer beams may be cut on the bias on the head end. If cut on the bias on the head end, the distance to the fold of the hinge strip shall be measured from the point of beam closest to the head end.

3.3.1.1 Alternate I-beam construction (class 1). The I-beam may be constructed by sewing a 1 1/4 inch (32 mm) wide strip of the coated specified in 3.1.3.2 to each edge of a 1-3/4 inch (44 mm) wide strip of the uncoated cloth specified in 3.1.1 as shown on drawing 2-3-97. Beginnings and ends of stitching shall be securely backstitched or tacked for 1/2 inch (13 mm). All breaks in stitching shall be overstitched for at least 1 inch (25 mm) back of break. The vertical raw edge of the uncoated nylon shall be fused to prevent fraying.

3.3.2 C-beam construction (class 2). All seams must be vulcanized full length. The C-beam shall be made from strips of either of the coated cloths specified in 3.1.3.1 or 3.1.3.2, not less than 2 1/2 inches (64 mm) wide. The heavily coated side of the cloth shall be molded to the top and bottom panels. The effective height of the partition shall be not less than 1 1/8 inches (29 mm). The C-beams shall be molded to the body of the mattress so that the C-beam material and the mold line shall be 1 1/8 ± 3/8 inches (29 ± 10 mm) from the inside junction of the fin edges, except the two outer beams shall be no closer than 3 1/2 ± 1/2 inches (89 ± 13 mm) from the inside junction of the fin edge at the head end of the mattress. The width of the mold line shall be not less than 1/4 inch (6 mm) on both top and bottom panels. The coated cloth shall be vulcanized after assembly of the mattress.

3.3.3 Cementing of batting. The batting shall be cemented to the inside or black side of the top mattress panel. A 2-inch (51 mm) wide continuous strip of the cement specified in 3.1.4 shall be applied to the black side of the top panel in the center of each channel for adhering the batting to the top panel. The cement strip in each channel shall be 62 ± 1 inches (1575 ± 25 mm) long. The side panels shall have the cement strip adjusted in the tapered portion so that the 2-inch (51 mm) wide strip of cement is centered in the tapered portion. The batting should be inserted in the channels in such a manner as to insure that the cement line is in the center of the batting and that the batting will not interfere with the adhesion of the beams or fin edges. There shall be no adhering of the batting to the bottom panel of the finished mattress. As an alternate to the above. The cement may be applied to the two inch wide area in a pattern consisting of a minimum of ten continuous rows of cement, each being not less than 1/16 inch wide. This pattern is minimum for spray gun application.

3.3.4 Fin edge.

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3.3.4.1 I-beam construction (class 1). A strip of the hinge material specified in 3.1.3.3, not less than 1 1/2 inches (38 mm) wide, folded as a hinge shall be full cemented to the top and bottom panels on the inside along the entire perimeter of the mattress. Top and bottom panels shall be cemented together forming a fin edge 3/8 to 3/4 inch (10 to 19 mm) in width.

3.3.4.2 C-beam construction (class 2). The top and bottom panels shall be molded together forming a fin edge 1/4 to 9/16 inch (6 to 14 mm) in width. A strip of gum stock may be inserted in the fin edge between the top and bottom panels providing the maximum overall weight requirement for the finished mattress is not exceeded.

3.4 Inflation test. The finished mattress shall be tested for inflation at 1.0 ± 0.1 pounds per square inch (psi) (6.9 ± 0.7 kPa) as specified in 4.3.3 with loss of pressure not to exceed 0.3 psi (2.1 kPa). There shall be no adherence of the top and bottom panels except where they are joined by the partitions and edge seams and there shall be no evidence of blown out beams. There shall be no evidence of cracking or other failure adversely affecting performance.

3.4.1 Low temperature inflation test. The finished mattress shall be subjected to the low temperature inflation test as specified in 4.3.3. Upon examination there shall be no evidence of cracking, leakage or other failure adversely affecting performance. The loss in pressure shall not exceed 0.5 psi (3.4 kPa).

3.5 Contractor's label. Each mattress shall be durably and legibly marked in black ink near one corner of the head end on the valve side in block letters approximately 1/2 inch (13 mm) high showing the following information:

Mattress, pneumatic, (insulated)
Class
Specification
Stock number
Contractor's name
Contract number
Month of manufacture
Lot number

3.6 Precautionary marking. Each mattress shall be durably and legibly stamped in black ink directly under inflating tube in letters 1/4 inch (6 mm) high with the following marking:

"Remove plug by pulling with fingers. Do not use cord to pull out plug. Inflate orally. Do not use mechanical pump or other high pressure method. Do not overinflate. Take care not to place the mattress on sharp objects that may abrade or puncture it. Test for proper inflation by sitting on mattress - Buttocks should barely touch the ground."

3.6.1 Use marking. Each mattress shall be durably and legibly stamped in black ink, approximately 1 inch (25 mm) below the inflation tube in letters 2 inches (51 mm) high with the letters "TOP." Directly under the letters "TOP", in letters 1/4 inch (6 mm) high the following marking shall be made:

"When inflated for use, always keep this side in contact with sleeping bag."

* 3.7 Weight. The weight of the finished mattress shall not exceed 3 pounds, 11 ounces.

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3.8 Repairs. Prior to the fabrication of the mattress, bare spots created by pickoffs or scuffs not exceeding 3 inches (76 mm) in any dimension may be touched up with the coating compound specified in 3.1.2. The repair compound shall be well adhered to the coated cloth. Each finished mattress may have one fully adhered cemented patch made from the cured coated cloth specified in 3.1.3.1 to repair a leak. The patch shall be not less than 1 inch (25 mm) or more than 2 inches (51 mm) in any linear dimension. The patch shall not extend over the fin edge. When subjected to scrubbing action between the hands, there shall be no lifting of the edge and no cracking, flaking, or removal of the repair compound or patch. Each repaired mattress shall pass the inflation test specified in 3.4.

3.9 Seam adhesion requirements. Cemented and molded seams shall conform to the following requirements when tested as specified in 4.3.3. The requirements are applicable to the required seam widths including seam tolerances:

	Adhesion strength, minimum			
	Before immersion		After immersion	
	Pounds	(newtons)	Pounds	(newtons)
Cemented seams (class 1):				
1 1/2 and 1 1/8 inches (38 and 29 mm) wide	4.0	18	4.0	18
1/2 inch (13 mm) wide	2.0	9	2.0	9
Molded seams (class 2):				
5/16 inch (8 mm) wide	3.0	13	3.0	13
1/4 inch (6 mm) wide	3.0	13	3.0	13

3.10 Marking legibility. The markings specified in 3.5 and 3.6 shall show fair fastness and retain legibility after testing for crocking as specified in 4.3.3.

3.11 Workmanship. The finished mattress shall be clean and well finished, and shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Inspection of components. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended modified, or qualified in this specification or applicable purchase documents. In addition, testing shall be performed on components and materials listed in table III for the characteristics shown. The sample unit for the base cloth and the cured coated cloth used for the top, bottom, and beam material (when applicable) shall be 5 yards (4.6 m) full width or equivalent panels in full width for each component with exceptions noted in table III. The sample unit for other components shall be as noted in table III. The lot size shall be expressed in units of 1 yard (0.9 m) full width for the base cloth and coated cloth and 1 gallon (3.8 liter) for cement. The methods of testing contained in FED-STD-191, whenever applicable, and as listed in table III shall be followed. The physical values specified apply to the average of determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The inspection level shall be S-1 and the acceptable quality level (AQL) shall be 4.0 expressed in defects per hundred units. For those characteristics where test results are reported as pass or fail, there shall be no evidence of failure of any test specimen to meet the requirements as specified.

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INSTRUCTIONS FOR TESTING — TABLE III

Component	CHARACTERISTIC	Specification Reference: Requirement	Specification Reference: Test Method	Requirements Applicable To: Individ Unit	Requirements Applicable To: Lot Aver	Number Determinations Per Unit	Results Reported As: Pass or Fail	Results Reported As: Numerically to Nearest	Inspect Level	AQL
Base cloth	Weave	3.1.1	Visual			1	X	-		
	Finish (heat set and not calendered)	3.1.1	1/			-	-	-		
	Identification of fiber	3.1.1	1530			1	X	-		
	Denier	3.1.1	1/			-	-	-		
	Filament	3.1.1	1/			-	-	-		
	Luster	3.1.1	1/			-	-	-		
	Weight	3.1.1	5041			5	-	0.1 ounce ($10\ g/m^2$)		
	Yarns per inch (yarns/cm)	3.1.1	5050			5	-	Whole number		
	Breaking strength (each direction)	3.1.1	5104			5	-	1 pound (1 N)		
	Tearing strength (each direction)	3.1.1	5134			5	-	0.1 pound (1 N)		
	Shrinkage	3.1.1	4.4.11			1	-	0.1 percent		
	Chloroform soluble material	3.1.1	2611			1	-	0.1 percent		
	pH value	3.1.1	2811			2	-	0.1 pH		
	Copper content	3.1.1	2053			2	-	.001 percent		
	Manganese content	3.1.1	2053			2	-	.0001 percent		
Coating compound	Composition	3.1.2	1/			-	-	-		

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INSTRUCTIONS FOR TESTING — TABLE III

CHARACTERISTIC		Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
Compound		Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Cured, coated cloth used for top and bottom pieces	Weight on lightly coated side	3.1.3.1	1/			-	-	-		
	Overall weight	3.1.3.1	5041			5	-	0.1 ounce (1 g/m^2)		
	Breaking strength (each direction)	3.1.3.1	5102			5	-	1 pound (1 N)		
	Tearing strength (each direction)	3.1.3.1	4.4.4			5	-	1 gram (1 N)		
	Hydrostatic resistance:									
	Initial	3.1.3.1	4.4.1			5	X	-		
	After accelerated aging	3.1.3.1	4.4.1.1			5	X	-		
	After weatherometer	3.1.3.1	4.4.1.2			5	X	-		
	After strength of coating	3.1.3.1	4.4.1.3			2	X	-		
	Adhesion of coating to cloth	3.1.3.1	4.4.2.1			3	-	0.1 lb (1 N) per 2 inch (51 mm) width		
	Stiffness at 70°F (21°C) 2/	3.1.3.1	4.4.3 & 4.4.3.1			5	-	1 mm		
	Stiffness at -40°F (-40°C) 2/	3.1.3.1	4.4.3 & 4.4.3.2			5	-	1 mm		
	Abrasion	3.1.3.1	4.4.8			3	X	-		
	Blocking	3.1.3.1	4.4.7			1	X	-		
	Resistance to cold crack	3.1.3.1	4.4.5			3	X	-		
	Resistance to accelerated aging	3.1.3.1	4.4.6			1	X	-		

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INSTRUCTIONS FOR TESTING — TABLE III

CHARACTERISTIC		Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
Component		Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Cured, single coated cloth for beams 3/ (when applicable)	Overall weight	3.1.3.2	5041			5	-	0.1 oz/sq yd (10 g/m^2)		
	Adhesion of coating to cloth	3.1.3.2	4.4.2.1			3	-	0.1 lb (1 N)		
Cured, coated cloth used for hinge strip (when required	Weight of coating on one side	3.1.3.3	1/			-	-	-		
Cement 4/	Composition	3.1.4 & 3.1.4.1	1/			-	-	-		
	Peel strength	3.1.4.1	ASTM Method D 903			1	-	0.1 lb per inch (1N/25 mm) of width		

1/ The contractor shall furnish a certificate certifying compliance to the applicable requirement.

2/ Sample unit shall be one 12 by 12 inch (305 by 305 mm) square of coated cloth cured as specified in 4.4.1.

3/ Sample unit shall be one yard (910 mm) full width.

4/ Sample unit shall be 1/2 pint (0.2 liter) composite and 1/2 yard (460 mm) of vulcanized coated cloth. Bond shall be made by cementing the heavily coated (black) sides of the two strips of each test specimen

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* 4.3.1.1 In-process inspection. Inspection shall be made of the vulcanization of the inflating tube and at any point during any phase of manufacture to assure compliance with all operations and processes which cannot be determined in the completely fabricated mattress. The Government reserves the right to exclude from consideration for acceptance any lot for which in-process inspection has indicated non-conformance.

4.3.2 Inspection of the end item.

4.3.2.1 Examinations. The defects found during the examination of the end item shall be classified in accordance with 4.3.2.1.1, 4.3.2.1.2 and 4.3.2.1.3. The lot size shall be the number of complete mattresses presented for acceptance. The applicable inspection levels and AQLs shall be as specified in 4.3.2.1.4.

4.3.2.1.1 Examination of the inflated end item. The sample unit for this examination shall be one complete mattress. The mattress shall be inflated to 1.0 ± 0.1 psi (6.9 ± 0.7 kPa).

Examine	Defect	Classification Major	Minor
Design and construction	Top not one piece	X	
	Bottom not one piece	X	
	Any component part missing	X	
Coating	Not evenly coated on outside with rubber compound		X
	Does not approximate Olive Green 207		X
	Color not uniform		X
	Peels or flakes from cloth	X	
	Not dull finish		X
	Any blister which leaves a bare spot	X	
	Any bare spot on unreinforced portion of mattress	X	
	Any bare spot on reinforced area, larger than 1/2 inch (13 mm).		X
	Imbedded foreign matter which, upon removal, leaves an uncoated spot	X	
NOTE:	Reinforced areas are seams, beams, chafing patch, valve tube, and hinge strip (when required).		

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Examine	Defect	Classification Major	Minor
Partitions	Any beam missing or misplaced more than 3 inches (76 mm) from the end of the mattress	X	
	Any loose or broken partition	X	
	Any adhesion between top and bottom sides of mattress more than 3 inches (76 mm) from the partition end of the mattress except where they are joined by partitions or side beams	X	
	Any portion of the partition less than 1/2 inch (13 mm) in height for a distance of 3 inches (76 mm) or more	X	
	More than 72 inches (1829 mm) of total partition length in which beam is less than 1 1/8 inches (29 mm) in height, but not less than 1/2 inch (13 mm) in height		X
	NOTE: A commercial caliper of suitable size with blunt ends shall be used in making measurements.		
	Broken bead on partition side of beam joining top and bottom panels	X	
	Contours of sections vary from Drawing 2-3-97	X	
	Wavy by more than 1/2 inch (13 mm) up to 36 inches (914 mm)		X
	Wavy over 36 inches (914 mm)	X	
Class 1	Not cemented along entire length and width of beam on both top and bottom	X	
	Blister (void indicating uncemented area) extending more than 1/2 inch (13 mm) in width dimension of beam	X	
	Blister extending more than 1/4 inch (6 mm) but not more than 1/2 inch (13 mm) in width dimension of beam		X
	Blister extending more than 3/4 inch (19 mm) in length dimension of beam	X	
	Blister extending more than 1/2 inch (13 mm) but not more than 3/4 inch (19 mm) in length dimension of beam		X

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Examine	Defect	Classification Major	Minor
Class 2	Not molded along entire length and width of beam on both top and bottom	X	
	Any blister in molded area	X	
Inflating tube and stopper	Any part missing	X	
	not black		X
	Not specified type and design	X	
	Excess flash, blistering, not properly trimmed		X
	Not located as specified	X	
	Not parallel to beams by more than 60°	X	
	Not parallel to beams by more than 30° but less than 60°		X
	Stopper or tube torn, cracked, mutilated, or distorted	X	
	Tube not fully cemented to mattress:		
	(a) Any wrinkle or delamination which extends from the edge of the cemented opening in the top panel to the rim of the inflating tube	X	
	(b) Any single unadhered area greater than 0.1 square inch (65 mm^2)	X	
	(c) Ten or more unadhered spots (up to 0.1 square inch (65 mm^2) per spot)	X	
	(d) More than 3 but less than 10 unadhered spots (up to 0.1 square inch (65 mm^2) per spot		X
	Tie hole missing or torn through	X	
	Tie hole not punched out cleanly	X	
	Any hole other than tie hole	X	
	Tube body not securely attached to mattress, i.e., flange of tube can be separated from mattress fabric with slight hand pressure	X	
	Excess gum or adhesive build up around valve		X
	Stopper not securely attached to tube by means of a cord		X
NOTE:	Suspend mattress by stopper. If cord detaches, it shall be considered a defect.		
	Color of cord not black		X
	Stopper does not fit snugly into tube	X	

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Examine	Defect	Classification Major	Minor
	Braid less than 11 1/2 inch (292 mm) or more than 12 1/2 inch (318 mm) in length		X
Fin edge	Missing	X	
	Not joined evenly		X
Class 1	Not cemented along entire perimeter of mattress	X	
	Uneven trim - variation between any two measurements is 3/4 inch (19 mm) or more	X	
	Width of fin edge less than 3/8 inch (10 mm)	X	
	Width of fin edge greater than 3/4 inch (19 mm)		X
Class 2	Not molded along entire perimeter of mattress	X	
	Any blister in molded area	X	
	Any open or unfilled channel in fin edge	X	
	Inside bead broken in any area	X	
	Width of fin edge less than 1/4 inch (6 mm), and 1 inch (25 mm) or more in length	X	
	Width of fin edge less than 1/4 inch (6 mm) and less than 1 inch (25 mm) in length		X
	Width of fin edge greater than 9/16 inch (14 mm)		X
NOTE:	Starting at the upper left hand corner of the mattress, measurements shall be taken at one foot (305 mm) intervals around the entire fin edge of the mattress.		
Workmanship	Any hole, tear, cut, crack, or window	X	
	Any evidence of leakage (check suspected area with soap and water solution or by immersing in water)	X	
	Not clean		X
	Mattress mutilated or distorted	X	
	More than four folds, punches, pleats or creases per mattress that cannot be corrected by manual pressure		X
	Objectionable odor		X
	Tacky surface	X	
	Any rubber or rubber coated part not properly vulcanized	X	

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Examine	Defect	Classification Major	Classification Minor
	Soiled; will stain clothing or other items which come in contact with the soiled area	X	
	Any touching up of bare spots that extends more than 3 inches (76 mm) in largest dimension		X
	Patch not neatly made		X
	Any part of patch not securely affixed to mattress, e.g., can be readily peeled with slight hand pressure	X	
	Two patches per mattress		X
	Three or more patches per mattress	X	
	Any patch over the fin edge	X	
	Linear dimension of patch more than 2 inches (51 mm), but less than 4 inches (102 mm) or less than 1 inch (25 mm) but not less than 3/4 inch (19 mm)		X
	Linear dimension of patch greater than 4 inches (101) mm) or less than 3/4 inch (19 mm)	X	
	Any pinch, pleat, crease, or fold of material fused together, where not intended, which results in any coating separation when it can be flattened out by application of manual pressure	X	
	Cemented opening in the top of panel through which inflating tube is inserted is more than 1 1/8 inches (28 mm) or less than 7/8 inch (22 mm)		X
Markings	Missing, incorrect, illegible, not type or size specified, not in proper location, or not accomplished in the specified manner		X
	Ink rubs off with moderate pressure of fingers		X
Inflated to 1/2 pound (3.5 kPa) (for warpage only)	Warp so that when placed on flat surface, base of any end section of mattress is raised:		
	- five inches (127 mm) or more from flat surface	X	
	- More than 3 inches (76 mm) but less than 5 inches (127 mm) from the flat surface		X

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4.3.2.1.2 Examination of the uninflated end item. The sample unit for this examination shall be one complete mattress.

Examine	Defect	Classification Major	Minor
Hinge strip (class 1 only)	Missing or detached	X	
	Not cemented along entire perimeter of mattress	X	
	Open channel extending from outside of hinge strip into fin edge	X	
	Blister (void, indicating uncemented area) extending more than 1/2 inch (13 mm) in width dimension of strip	X	
	Blister extending more than 1/4 inch (6 mm) but not more than 1/2 inch (13 mm) in width dimension of strip		X
	Blister extending more than 3/4 inch (19 mm) in length dimension of strip	X	
	Blister extending more than 1/2 inch (13 mm) but not more than 3/4 inch (19 mm) in length dimension of strip		X
Chafing patch	Missing	X	
	Not round		X
	Not located in required position opposite tube protrusion	X	
	Not firmly affixed	X	
	Less than 3 inches (76 mm) in diameter		X
	Not on inside of mattress		X
Batting	Missing	X	
	Not properly located		X
	Area of batting is loose or is not securely cemented to the top panel. (NOTE: When in doubt, check mattress after inflation)	X	
	Length less than 61 inches or more than 63 inches		X

4.3.2.1.3 Dimensional and weight examination of the end item. The sample unit for this examination shall be one uninflated mattress.

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Examine	Defect			
	More than		Less than	
	Inches	(mm)	Inches	(mm)
Overall length (less fin edge)	75 1/4	(1911)	73	(1854)
Distance between top and shoulder	13 1/4	(337)	12 1/2	(318)
Distance between shoulder and start of hip bevel	28 1/2	(724)	27 3/4	(705)
Distance between start of hip bevel and bottom	33 1/2	(851)	32 3/4	(832)
Width at top of mattress	18 1/2	(470)	17 1/2	(445)
Width of mattress (measure between shoulder and start of hip bevel)	33	(838)	30	(762)
Width at bottom of mattress	23 1/2	(597)	22 1/2	(572)
Distance between fin edge and each outer partition (measure at shoulder	7	(178)	6 1/2	(165)
Distance between each partition and adjacent partition	4 3/4	(121)	4 1/4	(108)
Distance between ends of beams to fold of hinge strip (class 1) or inner junction of fin edge (class 2) at end of mattress:				
- Three inner beams at both ends and 2 outer beams at bottom end	1 1/2	(38)	3/4	(19)
- Two outer beams at head end	4	(102)	3	(76)
Width of flange on each side of I-beams (class 1) (measure each side of top and bottom of beam)	-	-	1/2	(13)
Width of molded area on C-beam (class 2) (measure on both top and bottom pieces)	-	-	1/4	(6)
Weight of mattress	3 lbs 8 oz (1.6 kg)		-	-

4.3.2.1.4 Inspection level and AQL. The inspection levels and AQLs expressed in defects per hundred units shall be as follows:

	Inspection level	AQL
For examination in 4.3.2.1.1	II	2.5 major 6.5 total
For examination in 4.3.2.1.2	S-4	0.65 major 1.5 total
For examination in 4.3.2.1.3	S-3	6.5

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4.3.2.2 Packaging inspection. An examination shall be made to determine that the preservation-packaging, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; incomplete; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling Bulged or distorted container
Content	Number of mattresses per container is more or less than specified

4.3.2.3 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.3.3 Testing of the end item. The end item shall be tested for the characteristics listed in table IV. The sample unit shall be two mattresses and as necessary 1 yard (914 mm) of each seam type. The lot size shall be expressed in units of one mattress. The inspection level shall be S-2 for adhesion, low temperature inflation, ambient temperature inflation and marking fastness tests and for these four characteristics there shall be no evidence of failure to meet the requirements as specified. All test reports shall contain the individual values utilized in expressing the final result. Five mattresses from each established lot shall be tested for batting adherence tests. These mattresses shall be selected from those which were used for the low temperature inflation test. To the extent possible based on the sample size, the adhesion test specimens shall be cut from the mattress used in the batting adherence test. Where necessary, specially prepared test seams will be used. For class 1 mattresses, seams for test shall be prepared in the same way as mattress seams normally are prepared using the same amount of cement, rolling, tension and curing treatment as used in normal production. For class 2 mattresses, the test seams shall be molded using the same heat, pressure and time and shall be cured and cut in the same way as in normal production.

TABLE IV. End item tests

Characteristic	Requirement para	Test procedure	No. of determinations per sample unit	Results reported to nearest
Batting adherence	3.3.3	4.4.12	(see 4.3.3)	Pass or fail
Inflation	3.4	4.4.9	1	Pass or fail
Low temperature inflation	3.4.1	4.4.10	1	Pass or fail
Adhesion:				
Class 1 mattresses				
1 1/2 (38 mm) and 1 1/8 inch (28 mm) seams:				
-Initial	3.9	4.4.2.2.1	3	0.1 lb (1 N)
-After immersion	3.9	4.4.2.2.1	3	0.1 lb (1 N)
1/2 inch (13 mm) seams:				
-Initial	3.9	4.4.2.2.1	3	0.1 lb (1 N)
-After immersion	3.9	4.4.2.2.1	3	0.1 lb (1 N)
Class 2 mattresses				
5/16 inch (8 mm) seam (molded):				
-Initial	3.9	4.4.2.2.2	3	0.1 lb (1 N)
-After immersion	3.9	4.4.2.2.2	3	0.1 lb (1 N)
1/4 inch (6 mm) seam (molded):				
-Initial	3.9	4.4.2.2.2	3	0.1 lb (1 N)
-After immersion	3.9	4.4.2.2.2	3	0.1 lb (1 N)
Marking fastness	3.10	5651 of FED-STD-191	1	Pass or fail

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4.4 Test methods.

4.4.1 Hydrostatic resistance. The hydrostatic resistance test shall be conducted as specified in method 5512 of FED-STD-191 except that a 40 pound per square inch (276 kPa) water pressure shall be applied to the olive green coated side of the cloth and held for 1 minute. The specimen shall be observed for evidence of leakage.

4.4.1.1 After accelerated aging. After the cloth has been aged as specified in 4.4.6, the cloth shall be tested for hydrostatic resistance as specified in 4.4.1.

4.4.1.2 After weatherometer test. After the olive green coated side of the cloth has been exposed for 100 hours to the conditions specified in method 5804 of FED-STD-191, the cloth shall be tested for hydrostatic resistance as specified in 4.4.1.

4.4.1.3 After strength of coating. After the coated cloth has been stretched as specified in method 5972 of FED-STD 191, the cloth shall be tested for hydrostatic resistance as specified in 4.4.1.

4.4.2 Adhesion tests.

4.4.2.1 Coating adhesion. Coating adhesion shall be determined by method 5970 of FED-STD-191 except that the coating adhesion for both sides of the coated cloth shall be tested.

4.4.2.2 Seam adhesion.

4.4.2.2.1 Cemented seams, I-beam construction. Adhesion tests, wet and dry, shall be conducted as specified in method 5964 of FED-STD-191 except that the test report requirements shall not be applicable, The adhesion value shall be the average load registered for 3 inches (76 mm) of separation of the seam. The separation shall be between the top and bottom of the I-beam and the coated body cloth of the mattress. On the outside edge, the strapping shall be separated from the fin and two plies of the fin shall also be separated. A test for adhesion of plies of the I-beam (separation of plies) shall be conducted.

4.4.2.2.2 Molded seams, C-beam construction. Adhesion tests, wet and dry, shall be conducted as specified in method 5964 of FED-STD-191 except that the test report requirements shall not be applicable. The adhesion value shall be the average load registered for 3 inches (76 mm) of separation of the seam. The separation shall be between the top and bottom of the C-beam and the coated body cloth of the mattress. On the outside edge, the two plies of the fin shall be separated.

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4.4.3 Stiffness test. Stiffness tests shall be as specified in method 5204 of FED-STD-191 except that only the warp direction shall be tested. The specimens shall be cut from a square piece of coated cloth, 12 by 12 inches (305 by 305 mm), which has been cured under tension (to prevent curling) by the contractor at the same time-temperature cycle used for fabrication of the pneumatic mattress.

4.4.3.1 Stiffness at 70°F (21°C). The test specimens shall be tested at -40° ± 2°F (-40 ± 1°C) after conditioning at that temperature for not less than 5 hours. The test shall be conducted in still air.

4.4.3.2 Stiffness at -40°F (-40°C). The test specimens shall be tested at -40° ± 2°F (-40 ± 1°C) after conditioning at that temperature for not less than 5 hours. The test shall be conducted in still air.

4.4.4 Tearing test. Resistance to tearing tests shall be conducted as specified in method 5132 of FED-STD-191, using standard instrument with one augmenting weight.

4.4.5 Resistance to cold crack at -50°F (46°C). Resistance to cold cracking shall be conducted as specified in method 5874 of FED-STD-191. The test specimen shall be tested at -50 ± 5°F (-46 ± 3°C) after conditioning for not less than 5 hours at that temperature.

4.4.6 Accelerated aging test. A specimen of sufficient dimensions shall be exposed to the conditions specified in method 5852 of FED-STD-191 for a period of 4 days. The specimen shall be examined for cracking, flaking, peeling, stiffness, brittleness, softness, and tackiness. The breaking strength test shall not be performed.

4.4.7 Blocking test. Blocking tests shall be conducted as specified in method 5872 of FED-STD-191 except that one specimen from each sample unit shall be tested and the test temperature shall be 220° ± 1.8°F (104.4 ± 1°C).

4.4.8 Abrasion test. The lightly coated (olive-green) side of the cloth shall be tested as specified in method 5304 of FED-STD-191, using fine emery cloth and subjected to 20 continuous cycles under a tension of 6 pounds (27 N) and under a load of 3 pounds (13 N). Breaking strength shall not be determined.

4.4.9 Finished mattress inflation test. The finished mattress shall be inflated with air to a pressure of up to 1 psi (7 kPa) making certain that there is no adherence of the top and bottom panels except where they are joined by partitions or side seams. After permitting the mattress to stay inflated for approximately 4 hours to permit full extension of the mattress and absorption of air by the inner surface, the pressure shall be readjusted 1.0 ± 0.1 psi (6.9 ± 0.7 kPa) and left inflated for 16 hours ± 1 hour. The pressure shall then be remeasured. Loss in pressure shall not exceed 0.3 psi (2 kPa). The test shall be performed under ambient temperature conditions involving minimum variation at the beginning and end of the period. No correction shall be made to compensate for variations in temperature and no tests shall be made where such variations are greater than 10°F (6°C). Observe for evidence of cracking or other failure adversely affecting performance.

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4.4.10 Finished mattress low temperature inflation test. The mattress, which has been tested in 4.4.9, shall be rolled into a tight bundle and placed in a cold chamber at a temperature of -40°F (-40°C). After a period of 16 hours ± 1 hour, and while still at -40° ± 2°F (-40° ± 1°C), the mattress shall be inflated to a pressure of 1.0 ± 0.1 psi (6.9 ± 0.7 kPa) and left inflated at that temperature for 1 hour. The mattress shall then be examined for any failure adversely affecting performance. The pressure shall be remeasured and the loss in pressure shall not exceed 0.5 psi (3.5 kPa).

NOTE: Mattresses which have failed the test specified in 4.4.9 shall not be used for this test.

4.4.11 Shrinkage. A sample 20 inches (508 mcm) square shall be cut from the uncoated base cloth and conditioned for 4 hours in an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of 70° ± 5°F (21 ± 3°C). The sample shall be marked with an 18 inch (457 mm) square and formed into a loop by stapling two opposite sides together. A glass rod 1/4 inch (6 mm) in diameter and 21 inches (533 mm) long shall be placed inside the loop at the bottom. The sample shall be immersed in a tub of boiling water by attaching twine or wire to the ends of the top glass rod and allowing the sample to hang freely in the boiling water for 15 minutes. The sample shall be removed and allowed to drain. After the staples are removed, the sample shall be dried in a flat position on a horizontal screen and reconditioned as stated above. The marked square shall be measured in three places in the warp direction and three places in the filling direction and the amount of shrinkage determined.

4.4.12 Adherence of batting. Cross sections of the mattress shall be made at the head and foot of each mattress by cutting across the full width of the mattress. Each channel shall then be examined for non-adherence of the batting to the bottom of the mattress. Subsequent to this evaluation, each channel shall be cut in its full length for examination of adhesion of the batting to the top of mattress. Each channel shall have the batting adhered to the top of the mattress the full length of each batting strip. There shall be no evidence of failure to meet the requirements.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each mattress shall be laid out flat with the inflating tube side up. The width shall be folded in half. The top and bottom of the mattress shall be folded inward to meet in the center, The resulting length shall be folded in half. The folded mattress shall measure approximately 18 by 15 1/2 by 4 1/2 inches (460 by 390 by 110 mm). Care shall be exercised to see that all excess air is expelled during the process of folding. One folded mattress shall be placed in a snug-fitting fiberboard box conforming to style OPF, type CF (variety SW), class domestic, grade 125 of PPP-B-636. Each box shall be securely closed with 2 inch (51 mm) minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

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5.1.2 Level C. Each mattress shall be preserved-packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A packing. Five mattresses, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Mattresses, preserved-packaged level A, shall be packed on edge, five in length, one in width and one in depth within a shipping container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.2.2.

5.2.2 Level B packing. Five mattresses, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. Mattresses, preserved-packaged level A, shall be packed on edge, five in length, one in width and one in depth within a shipping container. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.2.2.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in appendix of the container specification, except that the inspection shall be in accordance with 4.3.2.2.

5.2.3 Level C packing. Mattresses, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Palletization. When specified (see 6.2), mattresses, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The insulated pneumatic mattress is intended for use in conjunction with arctic and mountain sleeping bags.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of the applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

c. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

d. When palletization is required (see 5.3).

6.3 Standard sample. Sample of standard shade olive green 207 coated cloth may be obtained from the procuring activity issuing the invitation for bids.

6.4 Supersession data. This specification includes the requirements of LP/P.DES 6-74 dated 15 February 1974, subject: Mattress, Pneumatic, (insulated).

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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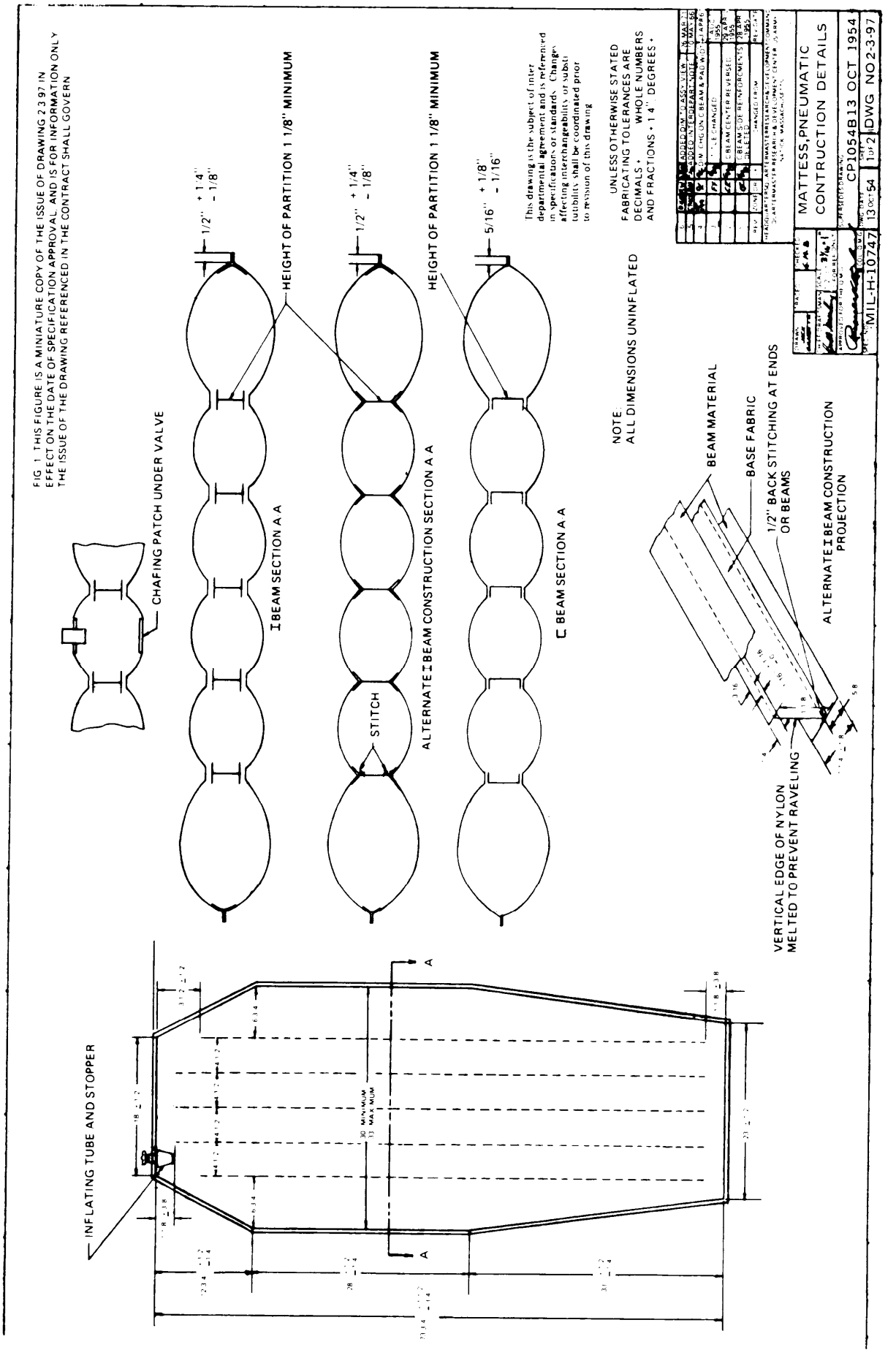
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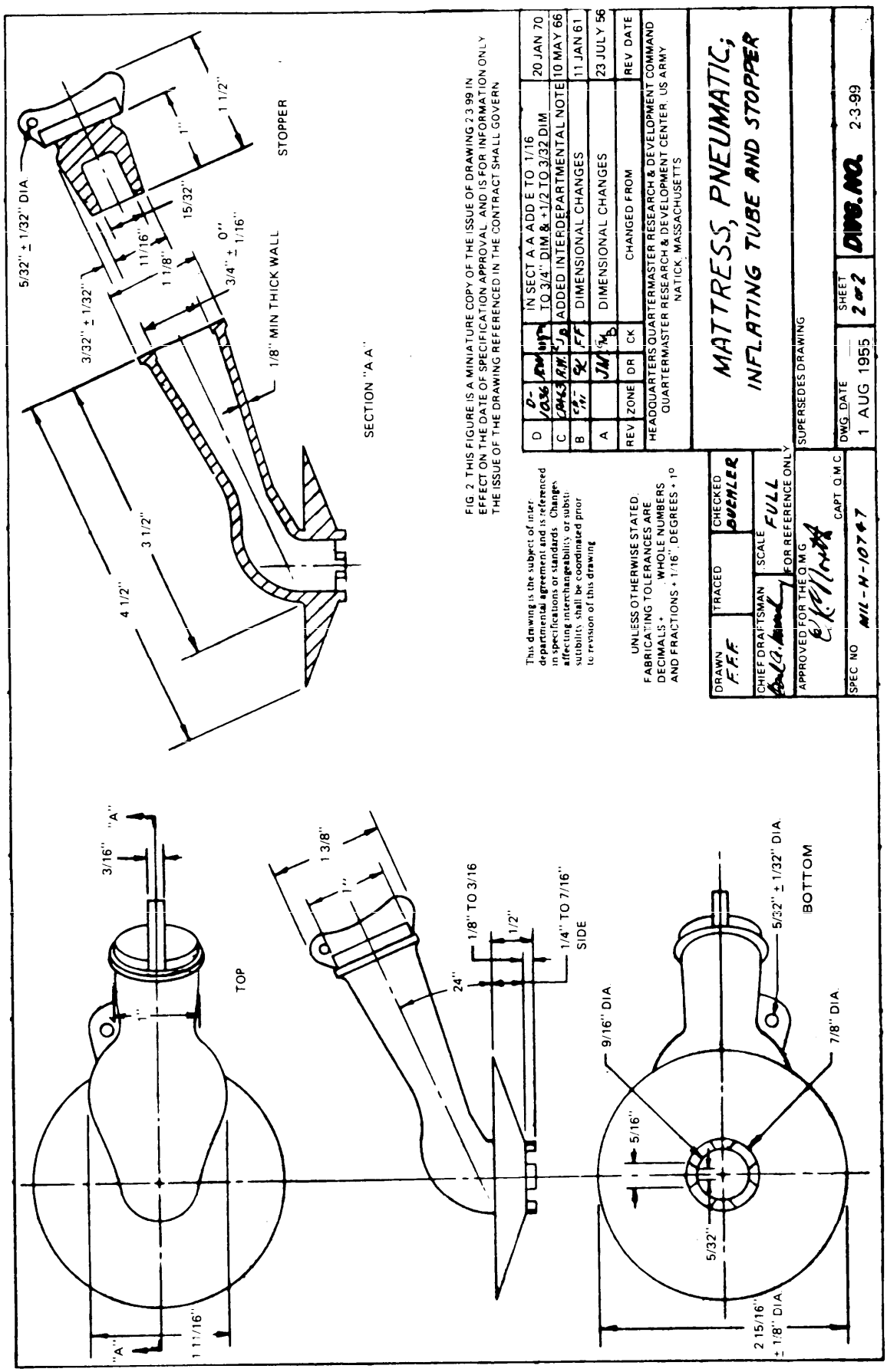


FIG 2 THIS FIGURE IS A MINIATURE COPY OF THE ISSUE OF DRAWING 2 3 99 IN EFFECT ON THE DATE OF SPECIFICATION APPROVAL AND IS FOR INFORMATION ONLY THE ISSUE OF THE DRAWING REFERENCED IN THE CONTRACT SHALL GOVERN

REV	ZONE	DR	CK	CHANGED FROM	REV DATE
D	0-136	RM	[illegible]	IN SECT A A ADD E TO 1/16 TO 3/4" DIM & +1/2 TO 3/32 DIM	20 JAN 70
C	0643	RM	JB	ADDED INTERDEPARTMENTAL NOTE	10 MAY 66
B	[illegible]	[illegible]	FF	DIMENSIONAL CHANGES	11 JAN 61
A		JM	MD	DIMENSIONAL CHANGES	23 JULY 56

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AND FRACTIONS ± 1/16", DEGREES ± 1°

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APPROVED FOR THE Q.M.G. E. K. Wilson CAPT. Q.M.C.

HEADQUARTERS QUARTERMASTER RESEARCH & DEVELOPMENT COMMAND
QUARTERMASTER RESEARCH & DEVELOPMENT CENTER, U.S. ARMY
NATICK, MASSACHUSETTS

MATTRESS, PNEUMATIC; INFLATING TUBE AND STOPPER

SUPERSEDES DRAWING

SPEC NO MIL-M-10747 DWG. DATE 1 AUG 1955 SHEET 2 of 2 DWG. NO. 2-3-99

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-43968A	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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